

Integer Linear Forms for a Factorial Reciprocal Series

Cancellation and attainable moments for Erdős Problem #68

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31 August 2026; revised 29 September 2026

ABSTRACT

We give integral coordinates for finitely supported integer vectors under factorial-weighted cancellation. Removing an auxiliary first coordinate reduces the attainable moments to one Diophantine equation. Its infinite gcd is determined by a finite interval ending below $2D^2$, where D is the cancellation depth. At depth four we use the coordinates to construct a vector with least positive moment 1380 and a nonintegral remainder. The remainder condition needed to deduce irrationality of $S = \sum_{n \geq 2} (n! - 1)^{-1}$ remains open.

1 Introduction

Erdős asked whether replacing $n!$ by $n! - 1$ preserves the irrationality of the reciprocal sum [1, p. 102]. The question appears as Problem 68 in Bloom’s catalogue [2].

Problem 1.1 (Erdős Problem #68). Is

$$S = \sum_{n \geq 2} \frac{1}{n! - 1}$$

irrational?

The usual proof for e compares a factorial multiple of the tail with an integer. For S , that multiplier need not clear the partial sum: already $3!H_3 = 36/5$, where $H_m = \sum_{n=2}^m (n! - 1)^{-1}$. We instead choose integral weights congruent to $n!$ modulo $d! - 1$. A finite integer combination of these weights gives a remainder that differs from MS by an integer, with M the corresponding factorial moment. We solve the equations that cancel its first weighted summands.

Theorem 2.1 describes every finite integer solution. An adjacent factorial difference affects only the divisors of its index; eliminating the proper divisors isolates one weighted sum. Two triangular changes with unit determinant make this an integral basis. Removing an auxiliary index-one coefficient then leaves one Diophantine equation.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Theorem 3.1 reduces its infinite gcd to a specified finite interval, giving the moment ideal and a vector attaining its generator.

These conclusions do not assume anything about S . Their application to irrationality requires one further property: for each proposed denominator $q > 0$, a vector with $q \mid M$ must have a nonintegral remainder. The depth-four example supplies this property for the divisors of 1380. Primitive vectors with moments eventually divisible by every fixed q are available, but their remainders are not known to be nonintegral.

Hančl and Tijdeman's tail-integrality lemma treats factorial series with integer coefficients [4, Lemma 2.1 and the following remark, p. 385]. Their factorial-scaled partial sums are integers. Our prefixes lack this property, so the integer-difference identity below replaces the clearing step. Koepf and Schmiersau's factorial-digit criterion [3, Example 3.2, p. 121] is used in the [companion's digit argument](#) to give an equivalent eventual-pattern condition for $S - e + 2$. Failure of that pattern at arbitrarily large indices also remains unproved.

Sections 2 and 3 give the coordinate construction and finite gcd. Section 4 uses them to construct and test a minimal-moment vector. Section 5 states the remaining nonintegrality question, with the secondary arithmetic criteria developed in the companion.

2 An integral basis

We seek integral weights congruent to $n!$ modulo $d! - 1$. For integers $d \geq 2$ and $n \geq 1$, take

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}}.$$

Write $n = kd + r$, where $0 \leq r < d$. The quotient is $r!$ times the multinomial coefficient $n! / ((d!)^k r!)$, and is therefore an integer. Since $n! = (d!)^k W_{d,n}$ and $d! \equiv 1 \pmod{d! - 1}$, it also satisfies $W_{d,n} \equiv n! \pmod{d! - 1}$. Thus replacing $n!$ by $W_{d,n}$ in a numerator changes the fraction with denominator $d! - 1$ by an integer. For instance, $W_{3,4} = 4$, and the change is $(24 - 4)/5 = 4$.

For a finitely supported integer vector $\lambda = (\lambda_n)_{n \geq 1}$, we define its factorial moment and weighted sums by

$$M(\lambda) = \sum_n \lambda_n n!, \quad V_d(\lambda) = \sum_n \lambda_n W_{d,n},$$

and let

$$\mathcal{R}(\lambda) = \sum_{d \geq 2} \frac{V_d(\lambda)}{d! - 1}.$$

Choose $N \geq 2$ beyond the support of λ . For $d > N$ the floor exponents are zero, so $V_d = M$ and the series converges absolutely. Only the first $N - 1$ terms can differ from those of MS , and the weight congruence gives

$$\mathcal{R}(\lambda) - M(\lambda)S = \sum_{d=2}^N \frac{V_d(\lambda) - M(\lambda)}{d! - 1} \in \mathbb{Z}. \quad (1)$$

For later use, $(n+1)! - 1 > (n+1)(n! - 1)$ gives the geometric tail bound

$$0 < S - H_N < \frac{N+2}{(N+1)((N+1)! - 1)} < \frac{2}{(N+1)! - 1} \quad (N \geq 2). \quad (2)$$

The majorant starts with $1/((N+1)! - 1)$ and has ratio $1/(N+2)$.

We now solve $V_2 = \dots = V_D = 0$ in integer vectors, allowing either sign of M . Write e_n for the unit vector at index n . The auxiliary coordinate at index one will make the basis triangular. It occurs only in the coefficient vector: the series still starts at $d = 2$, so no term $1/(1! - 1)$ is introduced. We impose $\lambda_1 = 0$ in Section 3.

To see the elimination in a small case, take $4e_3 - e_4$. Its factorial moment vanishes, and the only nonzero weighted sums are $V_2 = 6$ and $V_4 = 23$: the floor exponent changes only when d divides 4. Subtracting $6(2e_1 - e_2)$ removes V_2 without changing V_4 . We apply the same procedure successively to all proper divisors.

Theorem 2.1 (an integer basis with prescribed weighted sums). [Lean](#) · [Comparator](#) *Set*

[Lean](#)
[Comparator](#)

$$T_n = ne_{n-1} - e_n, \quad U_n = T_n - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2).$$

Then

$$M(U_n) = 0, \quad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \dots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \geq 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \quad (3)$$

Proof. First, $M(T_n) = n(n-1)! - n! = 0$. For a fixed d , the floor exponents at $n-1$ and n differ by one precisely when $d \mid n$. We therefore have

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d|n}.$$

Suppose the identities hold below n . For each proper divisor d , subtracting $W_{d,n}U_d$ cancels the weighted sum at d and changes none of the others. The weighted sum at n remains $n! - 1$, since $W_{n,n} = 1$. This proves the two identities by induction.

It remains to show that these vectors give all integer solutions. Restrict to indices $1, \dots, N$. The columns $e_1, T_2, \dots, T_N$ form a triangular integer matrix with diagonal $1, -1, \dots, -1$. Passing from the T_n to the U_n is another triangular integer change, now with diagonal one. Both matrices have determinant a unit, so their inverses have integer entries. We may therefore expand any integer vector uniquely in $e_1, U_2, \dots, U_N$. Applying M and V_d determines its coefficients as in (3). The coefficients with $d > N$ are zero because $V_d = M$. $\square$

For a prime $p \geq 3$ the divisor sum is empty, and $U_p = pe_{p-1} - e_p$. At a composite index the correction terms remove the extra weighted sums. For example,

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \quad (4)$$

Each U_n has moment zero and remainder one. We can thus alter an individual weighted sum by adding a multiple of U_n , while the fractional part of the remainder stays fixed. This will separate the cancellation equations from the eventual nonintegrality question.

3 Attainable moments

We first allow the auxiliary coordinate λ_1 and classify all vectors cancelling through a fixed depth $D \geq 2$. Put

$$L_D = \text{lcm}_{2 \leq d \leq D}(d! - 1), \quad K_D = L_D e_1 - \sum_{d=2}^D \frac{L_D}{d! - 1} U_d.$$

The integral coordinates in (3) imply

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1}. \quad (5)$$

When $V_2 = \dots = V_D = 0$, these congruences force $L_D \mid M$. Substituting $M = tL_D$ into the basis expansion gives

$$V_2 = \dots = V_D = 0 \iff \lambda = tK_D + \sum_{n>D} z_n U_n, \quad M = tL_D, \quad (6)$$

where the integers z_n have finite support. Since $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$, their remainders are

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n, \quad H_D = \sum_{d=2}^D \frac{1}{d! - 1}. \quad (7)$$

We now require support on $n \geq 2$. Write $a_D = [e_1]K_D$ and $u_n = [e_1]U_n$ for the first coordinates. Setting $\lambda_1 = 0$ gives

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (8)$$

Let $g_D = \gcd\{u_n : n > D\}$. Since the z_n may be arbitrary integers of finite support, their sums $\sum z_n u_n$ form the ideal $g_D \mathbb{Z}$. The moments of the admissible vectors are consequently

$$\mu_D \mathbb{Z}, \quad \mu_D = L_D \frac{g_D}{\gcd(g_D, a_D)}. \quad (9)$$

Indeed, (8) has a solution exactly when $g_D \mid ta_D$, or equivalently when $g_D / \gcd(g_D, a_D)$ divides t . Bézout's identity uses finitely many u_n , so the generator is attained by a finite vector. We show next that $g_D > 0$. Any vector attaining μ_D is primitive: dividing its coefficients by a common factor greater than one would preserve the cancellations and give a smaller positive moment.

3.1 A finite gcd calculation

The recursion for the first coordinate is

$$u_2 = 2, \quad u_n = - \sum_{\substack{d \mid n \\ 2 \leq d < n}} W_{d,n} u_d \quad (n > 2).$$

Induction gives $u_n = 0$ at odd indices. At twice a prime it gives $u_{2p} = -(2p)!/2^{p-1} \neq 0$, including $p = 2$, so the tail contains a nonzero coefficient and $g_D > 0$. To compute this infinite gcd, we show that a specified finite interval already forces divisibility of every later coefficient.

Theorem 3.1 (a finite formula for the gcd). [Lean](#) · [Comparator](#) Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then

[Lean](#)
[Comparator](#)

$$g_D = \gcd(u_{D+1}, \dots, u_H), \quad H < 2D^2.$$

Proof. Let $g = \gcd(u_{D+1}, \dots, u_H)$. The interval contains 2ℓ , so $u_{2\ell} = -(2\ell)!/2^{\ell-1}$ shows that $g > 0$ and $g \mid (2\ell)!$. We prove by strong induction that $g \mid u_n$ for all $n > D$.

The assertion holds through H by definition. For $n > H$ split the proper divisors in the recurrence at D . If $d \leq D$ and $d \mid n$, then $n/d \geq 2\ell$. We also have

$$(n/d)! \mid W_{d,n} \quad (d \mid n),$$

because the quotient counts partitions into n/d unordered blocks of size d . Hence $(2\ell)! \mid W_{d,n}$, and therefore $g \mid W_{d,n}$, for each small divisor. If $D < d < n$, the induction hypothesis instead gives $g \mid u_d$. Every summand of the recurrence is divisible by g , completing the induction. The finite gcd thus divides the whole tail, while the reverse divisibility follows from inclusion of the finite interval. Finally, $H \leq D(2D - 1) < 2D^2$. $\square$

Bertrand's postulate supplies the prime ℓ when $D \geq 3$; for $D = 2$ take $\ell = 2$. The proof requires $n/d \geq 2\ell$ for every small divisor, which is why it uses the horizon $H = D(2\ell - 1)$. A truncation at $2D$ is not justified by this argument.

The finite gcd also gives a construction. Compute integers b_n with $\sum_{n=D+1}^H b_n u_n = g_D$ by the extended Euclidean algorithm, and put $h = \gcd(g_D, a_D)$. Taking

$$t = g_D/h, \quad z_n = -(a_D/h)b_n \quad (D < n \leq H)$$

makes $ta_D + \sum z_n u_n = 0$. Substitution in (6) therefore produces a vector with moment μ_D . The same finite calculation proves minimality and supplies an attaining vector. Additional support or coefficient-norm requirements are separate optimisation problems.

4 A vector of least positive moment

At depth four the recursion gives

$$L_4 = 115, \quad K_4 = -55e_1 + 16e_2 + 3e_3 + 5e_4, \quad a_4 = -55.$$

Take $\ell = 3$ in Theorem 3.1, so that $H = 20$. The coefficients in this finite interval have gcd 60. There is a useful short proof: $u_6 = -180$, $u_8 = -4200$ and $23u_6 - u_8 = 60$, while

$$\text{lcm}(1, \dots, n) \mid u_n \tag{10}$$

gives the reverse divisibility for every $n > 4$. To prove (10), put $\Lambda_n = \text{lcm}(1, \dots, n)$. For a prime r , write v_r for its valuation. If $d \mid n$, Legendre's formula gives

$$v_r(W_{d,n}) = \sum_{j \geq 1} \left(\left\lfloor \frac{n}{r^j} \right\rfloor - \frac{n}{d} \left\lfloor \frac{d}{r^j} \right\rfloor \right) \geq v_r(\Lambda_n) - v_r(\Lambda_d).$$

Each summand is nonnegative, and each power $d < r^j \leq n$ contributes at least one. Hence $\Lambda_n \mid W_{d,n}\Lambda_d$, and induction in the recurrence, starting from $u_2 = 2$, proves the assertion. Formula (9) now gives

$$\mu_4 = 115 \frac{60}{\gcd(60, 55)} = 1380.$$

We construct a vector with this moment by taking $t = 12$ in (8). The choice $z_6 = -27$, $z_8 = 1$ works because

$$12(-55) - 27(-180) - 4200 = 0.$$

The vector $12K_4 - 27U_6 + U_8$ therefore has no index-one coefficient and cancels V_2, V_3, V_4 . To remove its coefficients at indices four and seven, add $-26U_5 + 8U_7$. These corrections still cancel through four and do not restore index one, since $u_5 = u_7 = 0$. Thus

$$\lambda = 12K_4 - 26U_5 - 27U_6 + 8U_7 + U_8,$$

which expands to

$$\lambda = 1482e_2 - 784e_3 - 136e_5 + 83e_6 - e_8, \quad (M, V_2, V_3, V_4) = (1380, 0, 0, 0). \quad (11)$$

Table 1 verifies all four sums directly.

n	λ_n	$n!$	$W_{2,n}$	$W_{3,n}$	$W_{4,n}$
2	1482	2	1	2	2
3	-784	6	3	1	6
5	-136	120	30	20	5
6	83	720	90	20	30
8	-1	40320	2520	1120	70

Table 1: The depth-four vector. Multiplying the last four columns by λ_n and summing gives 1380, 0, 0, 0.

The moment and all cancellations follow from the coordinates before expansion. Moreover, the integer correction to the remainder in (7) is the sum $-26 - 27 + 8 + 1 = -44$. Thus

$$\mathcal{R}(\lambda) = 1380(S - H_4) - 44.$$

Exact rational arithmetic gives

$$-31 + \frac{4}{5} < 1380 \sum_{d=5}^8 \frac{1}{d! - 1} - 44 < -31 + \frac{5}{6}, \quad \frac{2 \cdot 1380}{9! - 1} < \frac{1}{100}.$$

The positive remaining tail is less than $2/(9! - 1)$, so $-31 < \mathcal{R}(\lambda) < -30$. If $S = a/q$ with $q \mid 1380$, (1) would make this remainder an integer. Every divisor of 1380 is therefore excluded.

The companion gives the [alternative dual congruence, support and coefficient-norm refinements](#). They concern different optimisation questions. Minimality of the moment here follows from the ideal formula, without requiring a sparsest vector.

5 Nonintegrality of the remainder

Fix a vector with $V_2 = \dots = V_D = 0$ and $M > 0$. We may choose the cutoff $N \geq D$ anywhere at or beyond its support. Separate the finite signed sum from the positive tail:

$$A_N = \sum_{d=D+1}^N \frac{V_d(\lambda)}{d! - 1}, \quad \mathcal{R}(\lambda) = A_N + M(S - H_N).$$

By (2), $0 < S - H_N < 2/((N+1)! - 1)$. We can prove nonintegrality by comparing its upper bound with the distance from A_N to the next integer:

$$\frac{2M}{(N+1)! - 1} < \lfloor A_N \rfloor + 1 - A_N \quad (12)$$

Under this inequality, $\mathcal{R}(\lambda)$ lies strictly between consecutive integers, excluding every denominator dividing M . The argument allows either sign of A_N . Its strict gap is one when A_N is integral and can be arbitrarily small when A_N approaches an integer from below.

There are primitive solutions with $0 < M < N!$ and moments eventually divisible by any prescribed q . The [progression construction in the companion](#) gives such a family by turning $V_d = 0$ into roots of a polynomial. At its support endpoint the omitted tail is less than $2/N$, but the gap on the right of (12) may shrink too. Irrationality requires the comparison for a vector whose moment is divisible by each prescribed q , not merely an unbounded sequence of moments. No minimal-support assertion is made for this family.

Increasing the cutoff is a different operation from choosing a new vector. For $-6e_2 + e_4$ the support ends at four, where the test fails because $9/115 < 24/119$. At $N = 5$ it succeeds because $13376/13685 > 24/719$. More generally, for a fixed vector with $M > 0$, the quantities A_N increase to $\mathcal{R}$. If this limit is nonintegral, their gaps to the next integer tend to a positive number, so the test eventually succeeds. If the limit is integral, the eventual gap equals the omitted tail and the test fails. Thus extending the cutoff eventually detects a nonintegral remainder, but leaves both the remainder and the moment unchanged.

Moment-zero corrections also have a precise limitation. Each U_n changes the remainder by one. More generally, (1) shows that two vectors with the same moment have remainders differing by an integer. Such changes can simplify support and weighted sums, as they did in the example, but cannot change whether the remainder is integral.

The companion treats the [equivalent carry condition](#) and the [complementary-denominator tail inequality](#). It also records two larger finite computations: a non-unit carry at $m = 300000$ excludes every $q \mid 299999!$, and a continued-fraction enclosure excludes every $q < 2^{39990}$. Both computations were performed outside Lean; their [algorithms](#), [exact outputs](#) and [scope](#) are given there. The factorial-divisibility exclusion does not exclude all denominators with only small prime factors, whose exponents may be larger. Neither finite restriction establishes the required condition at arbitrarily large indices.

6 Verification

Formal proofs. A result with a kernel-checked Lean proof carries links beside its heading and in the margin; both open the same evidence. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. Inline citations to individual declarations elsewhere may identify ingredients of a proof. *Comparator* opens the recorded comparison, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean’s Comparator tool, allowing only the three standard axioms. *Comparator pending* means that the complete statement has no recorded comparison yet. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. Comparator checks statements, axioms and kernel acceptance; it does not establish novelty or peer review. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text.

The integral-coordinate theorem and finite-horizon gcd formula have recorded exact Lean bindings. The [companion source concordance](#) identifies their declarations and the attainable-moment theorem. Its large finite computations are separate from these kernel proofs.

Acknowledgements

The author thanks Wouter van Doorn for advice on exposition: explaining notation when it first appears, avoiding private terminology, and saying how restrictive a conditional hypothesis is. His advice concerned the writing of another note; he has not reviewed the mathematics of this paper. An AI research pass supplied by Will Cook derived the depth-four dual congruence and shorter vectors from the factorial-channel definitions and general moment-ideal theory developed earlier here. OpenAI Codex checked and integrated them; a separate AI pass reviewed the proof.

References

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- [4] Jaroslav Hančl and Robert Tijdeman. [On the irrationality of factorial series](#). *Acta Arithmetica* **118** (4) (2005), 383–401.